

Blockchain-Based Secure Communication of Intelligent Transportation Digital Twins System

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Abstract—The present work aims to improve the communication security of Internet of Vehicles (IoV) nodes in intelligent transportation through studying the safety of IoV in smart transportation based on Blockchain (BC). An IoV DTs model is built by combining big data with Digital Twins (DTs). Then, regarding the current IoV communication security issues, a secure communication architecture for the IoV system is proposed based on the immutable and trackable BC data. Besides, Wasserstein Distance Based Generative Adversarial Network (WGAN) model constructs the IoV node risk forecast model. Because the WGAN model calculates the loss function through Wasserstein distance, the learning rate of the model accelerates remarkably. After ten iterations, the loss rate of the WGAN model is close to zero. Massive in-vehicle devices in IoV are connected simultaneously to the base station, causing network channel congestion. Therefore, a Group Authentication and Privacy-preserving (GAP) scheme is put forward. As users increase during authentication, the GAP scheme performs better than other authentication access schemes. In summary, the Intelligent Transportation System driven by DTs can promote intelligent transportation management. Besides, introducing BC into IoV can improve access control's accuracy and response efficiency. The research reported here has significant value for improving the security of the information sharing of the IoV.

Index Terms—Internet of Vehicles, digital twins, blockchain, secure communication, intelligent transportation.

I. INTRODUCTION

NOW, the transportation systems are enormous enough. Improving travel experiences requires state-of-art,

intelligent, and digital approaches, such as autonomous driving and vehicle-road coordination, to accelerate the smart upgrade of transportation systems [1]–[3]. The Internet of Vehicles (IoV) originates from the Internet of Things. IoV perceives information about moving vehicles. It connects vehicles, pedestrians, and service platforms using the new generation of information and communication technologies. Hence, vehicles become intelligent, providing users with secure, comfortable, and intelligent traveling experiences and transportation services while improving efficiency [4]. It is often discussed together with autonomous driving or advanced driving aids. Ideally, all vehicles in IoV are equipped with an automatic driving system. This combination can aid drivers, improve transportation efficiency, and reduce traffic accidents. IoV monitors road information in real-time using the Global Positioning System and plans the most suitable route for the drivers. The intelligent voice system can achieve barrier-free communication between vehicles and humans. Intelligent sensing devices can promptly perceive the vehicles' environment while driving, avoiding accidents in time. Emerging technologies, such as Artificial Intelligence (AI), Blockchain (BC), and Digital Twins (DTs), are gradually applied to IoV. In this way, IoV promotes the deep integration of transportation and automobile industries, boosting intelligent transportation and autonomous driving techniques.

DTs have been applied to spatial architecture, including intelligent buildings, parks, and cities [5]. Afterward, with the transformation and upgrading of traditional industrial enterprises, DTs have been extended to smart manufacturing under Industry 4.0. The fifth-generation (5G) IoV and IoV industrial chain is combined with core DTs techniques to construct a new digital transportation infrastructure [6], [7]. DTs in IoV build driver behavior models and forecast driver behaviors under different road conditions based on historical driving data. Meanwhile, sharing these behavior models with other vehicles can accurately forecast dangerous situations, improving driving security.

IoV provides convenience for transportation management via information sharing between sensor nodes. However, it is vulnerable to security attacks. For example, wireless communication between IoV nodes and frequent network topology changes make the network vulnerable to numerous attacks. As a shared distributed ledger, BC applies to various links in IoV because of its non-tampering, non-deleting, openness, and transparency features [8]–[10]. Information collected by IoV will be uploaded to the BC platform, forming a decentralized and distributed big data sharing market. With drivers'

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authorization, the data will be applied to intelligent transportation, insurance, and automobile markets.

The IoV security based on DTs aims to improve the service quality of the intelligent transportation system (ITS) and strengthen the security of information sharing. First, DTs modeling empowered by the Cyber-Physical System (CPS) in IoV is explored. On this basis, security risks in IoV communication are analyzed. A BC-based secure communication framework is proposed to ensure the confidentiality of node information regarding the possible attacks on nodes during information accessing. A BC consensus mechanism matching the IoV characteristics is designed, enhancing data communication security. Here, a BC-based IoV security communication framework is constructed via the decentralization nature of BC. Besides, the corresponding access control mechanism is proposed. It can effectively enhance the confidentiality, integrity, and availability of information resources. This is of great significance to strengthen the security of IoV information sharing.

II. RELATED WORK

The development of IoV realizes the sharing of massive real-time information generated by vehicles among different participants. DTs models vehicles from different angles based on historical data. It can also update the model parameters using the new data to present the vehicle status in real-time intuitively. Zhang *et al.* [11] dynamically simulated the DTs model of mobile Electric Vehicles (EVs) and charging facilities. This model could accurately assess the performance of different IoV navigation algorithms. Interactions between IoV charging facilities and EVs were simulated by introducing a mobility model with DTs. Bhatti *et al.* [12] researched autonomous navigation control via DTs, vehicle wellness monitoring, and electric vehicle drives in intelligent vehicle systems. They built a complicated DTs system for EVs and sorted out the detailed workflow, benefiting the IoV management.

IoV can make transportation management intelligent; meanwhile, IoV communication security is also a hot topic. Jiang *et al.* [13] proved that BC showed exceptional advantages as a powerful approach for distributed storage and security management. On this basis, they applied BC to IoV's big data system and constructed a vehicle BC data outward transmission model. Kang *et al.* [14] built an interaction model between active and standby miners based on BC and contract theory to ensure safe data sharing. Wang *et al.* [15] proposed a practical distributed IoV authentication mechanism based on the BC consensus algorithm. Moreover, simulations under the veins framework verified the scheme's feasibility in reducing selfish behaviors and hostile attacks.

Intermediaries may tamper with communication data in the IoV data communication process between IoV nodes. Consequently, the confidentiality and integrity of information will decrease. The existing solution is to encrypt the communication data between the communication parties through symmetric encryption or asymmetric encryption. Common symmetric encryption algorithms include Data Encryption Standard and Advanced Encryption Standard. Both parties share the same encryption and decryption keys based on symmetric encryption

in data transmission. Symmetric encryption has the advantages of small amounts of calculation and fast speed. It is suitable for encryption scenarios with high real-time requirements and multitudes of data. Common asymmetric encryption algorithms include ElGamal and Elliptic curve cryptography. The key is public, and the node only needs to keep its private key to ensure the security of data transmission. Therefore, asymmetric encryption has the advantage of heightened security. Alaya and Sellami [16] introduced an efficient key management scheme based on symmetric and asymmetric encryption. This scheme considered the location, speed, direction, and the number of adjacent vehicle nodes. It effectively improved the information transmission rate of the Vehicle Ad Hoc Network. La manna *et al.* [17] applied an attribute-based encryption scheme in hardware/software platforms for automobiles. These platforms could improve the security of wireless update functions in the automobile scene. Moreover, they verified the effectiveness of the scheme through experiments.

To sum up, DTs in IoV focus on reducing traffic flow based on actual traffic data and providing a decision basis for various scheduling problems. Still, scholars rarely deeply research DTs modeling specifications under the IoV and CPS environment. Specific DTs modeling approaches are analyzed in the present work to fill in this gap. Furthermore, applying BC to IoV can ensure data transmission security via encryption technology. A security scheme for data transmission with asymmetric encryption is proposed to ensure data security. Meanwhile, it can reduce the time consumption of data encryption and decryption.

III. SECURITY OF COMMUNICATION OF VEHICLE NETWORK BASED ON BC TECHNOLOGY

A. 6G DTs-Driven IoV Management

In the 5G era, the International Telecommunication Union has identified three application scenarios. They are enhanced Mobile BroadBand (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC). The 6G-oriented application is expanded based on original application scenarios to cover all MBB scenarios gradually.

The critical capabilities of 5G communication have been improved considerably compared with current IoV communication techniques. The transmission delay of 5G networks can be as slight as milliseconds. This breakthrough meets the stringent requirements of IoV and ensures the safety of vehicles at high speed. 5G peak throughput reaches 10-20Gbit/s, and the connection density attains 1 million/km². This performance satisfies the communication demands among cars, humans, and transportation infrastructure in the future IoV environment. 6G can improve network performance significantly based on 5G. 6G peak throughput is more than 1Tbit/s, improving the network performance by more than 50 times. Moreover, the connection density is more than 10 million/km². The user experience data rate also attains 1Gbit/s.

Vehicle to Everything (V2X) needs to determine business scenarios and requirements. 6G V2X business scenarios comprise four groups [18], [19]. (1) Vehicle formation guarantees

that vehicles can travel together in a dynamic formation. All vehicles obtain information from the leader to manage the formation. (2) Extended sensors achieve local sensor information and real-time video images exchanged among vehicles, roads, drivers, and V2X servers. (3) Advanced driving can support automatic driving. Sharing the vehicle's sensor data can synchronize information among multiple vehicles and coordinate their driving trajectories. (4) Remote driving enables a remote driver or IoV application server to control the vehicle remotely. URLLC is the principal requirement for remote driving.

AI approaches in 6G scenarios are profoundly integrated with edge computing to form edge intelligence in a multi-level environment. This integrated technique can support dynamic and distributed collaborative computing services between vehicles and roads. Moreover, DTs map physical entities into the virtual world using digital means, providing the real-time interaction of agents in virtual and physical spaces. DTs technology adopts the global mapping model constructed by actual traffic data. It promotes intelligent traffic management based on real-time perception, multi-modal terminals, and multi-source data fusion of 6G networks.

B. Modeling Big Data-Based DTs

Cyber-physical integration is a necessary condition for intelligent manufacturing. CPS is the preferred approach for cyber-physical integration. It integrates the information network world and the dynamic physical world [20]. Due to tight connections and feedback loops, the physical and computational processes are excessively interdependent. CPS implements computation and communication closely interacting with the physical process. These operations essentially aim to add new functions to the physical system. CPS emphasizes the powerful computing and communication capabilities of the information world, improving the accuracy and efficiency of the physical world [21], [22].

DTs map the physical model to the digital model to exchange data. In this way, the performance of the physical entity can be optimized continuously. A DTs model of IoV is built by combining big data with DTs. Figure 1 illustrates the overall structure of the DTs model of IoV.

The physical entity layer transmits real-time data collected by sensors to the cloud network layer in the DTs model. The physical entity layer contains multiple physical entities such as components and equipment. DTs create virtual entities corresponding to physical items and interact with them to provide intelligent services [23]–[25]. The cloud network layer is responsible for modeling management and model simulation. Modeling management involves digital modeling and physical object visualization. The storage resource pool provides storage areas for physical entities and DTs to store their state information. Several DTs communicate mutually in the information communication layer, forming an information communication group. These information communication groups can communicate with each other as well. The system application layer takes charge of data mining, information processing, and operation maintenance analysis. The final user

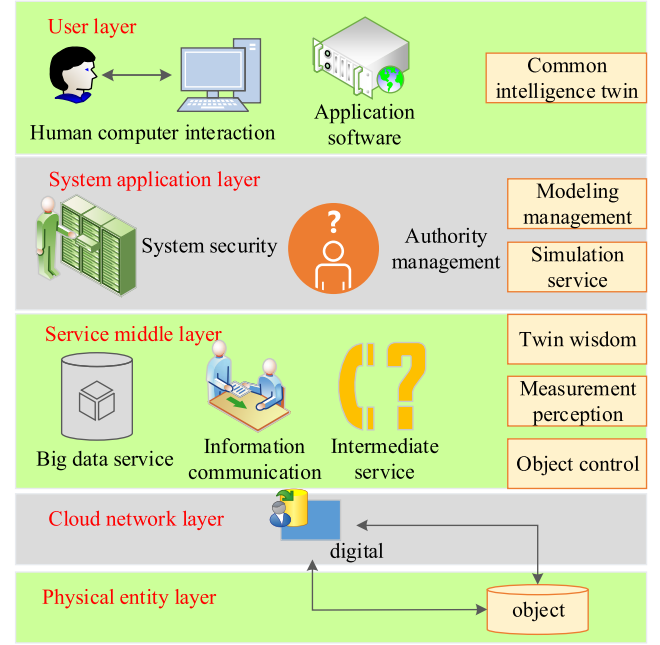


Fig. 1. Big data-based DTs model.

TABLE I
PARAMETER SUMMARY LIST OF THE DTs DATA
PROCESSING MODEL BASED ON SVM

λ	regularization parameter
α	LaGrange multiplier
I	identity matrix
K	$m \times m$ kernel matrix
D	the distance between the reporting location and the base station
β_s, β_r	preset parameter
T_{cer}	the difference between the time of reporting and certificate issuance

layer includes Human-Computer Interaction (HCI) and sharing twins, visualizing the information via HCI.

Two DTs layer data processing modes and a forecast model are proposed using Support Vector Machine (SVM) based on IoV. The data processing mode of the physical layer is suitable for scenarios with requirements for real-time performance. On the contrary, the data processing mode of the DTs layer is ideal for scenarios with a need for high computing power. Suppose there are m training samples. In that case, $\{(x_i, y_i)\}_{i=1}^m$ represents the input vector of the n -dimensional training sample. The segmentation hyperplane $WX + b = 0$ is searched for sample classification. Table I shows the related parameter symbols.

Hence, SVM optimization can be converted into maximizing the geometric interval of training samples, as shown in Equation (1).

$$\max_{W, b} \frac{r}{\|W\|} \quad (1)$$

$$s.t., y_i (W \cdot x_i + b) \geq r, i = 1, 2 \dots m \quad (2)$$

Setting the training samples' interval to 1 can get:

$$\max_{W, b} \frac{1}{2} \|W\|^2 \quad (3)$$

$$s.t., y_i (W \cdot x_i + b) \geq 1, i = 1, 2 \dots m \quad (4)$$

The Least Squares Support Vector Machine (LSSVM) implements self-adjustment of data processing in the DTs model. LSSVM introduces an error variable e_i into each training sample. This variable can solve the partial peculiarities of SVM and performs L2 regularization on itself simultaneously. LSSVM optimization can be described as:

$$\max_{W,b} \frac{1}{2} \|W\|^2 + \frac{\lambda}{2} \sum_{i=1}^m e_i^2 \quad (5)$$

$$s.t., y_i (W \cdot x_i + b) = 1 - e_i, i = 1, 2 \dots m \quad (6)$$

where λ refers to the regularization parameter.

LSSVM optimization is a problem with equality constraints. The LaGrange function can be obtained by introducing the LaGrange multiplier α_i :

$$L(W, b, e, \alpha) = \frac{1}{2} \|W\|^2 + \frac{\lambda}{2} \sum_{i=1}^m e_i^2 - \sum_{i=1}^m \alpha_i \{ [y_i (W \cdot \varphi(x_i) + b)] - 1 + e_i \} \quad (7)$$

In Equation (7), the function $\varphi(x_i)$ aims to map x_i into a higher-dimensional space.

Assuming the derivative to be 0, the partial derivative of each function variable is calculated according to:

$$\frac{\partial L}{\partial W} = 0 \rightarrow W = \sum_{i=1}^N \alpha_i \varphi(x_i) \quad (8)$$

$$\frac{\partial L}{\partial b} = 0 \rightarrow \sum_{i=1}^N \alpha_i = 0 \quad (9)$$

$$\frac{\partial L}{\partial e_i} = 0 \rightarrow \alpha_i = \lambda e_i \quad (10)$$

$$\frac{\partial L}{\partial \alpha_i} = 0 \rightarrow W \cdot \varphi(x_i) + b + e_i - y_i = 0 \quad (11)$$

α and b are calculated according to Equation (12).

$$\begin{bmatrix} 0 & \vec{1}^T \\ \vec{1} & K + \frac{\lambda}{I} \end{bmatrix} \begin{bmatrix} b \\ \alpha \end{bmatrix} = \begin{bmatrix} 0 \\ y \end{bmatrix} \quad (12)$$

In Equation (12), I represents the identity matrix, and K denotes the $m \times m$ kernel matrix. LSSVM's output for a new sample x is:

$$f(x) = W \cdot \varphi(x) + b = \sum_{i=1}^m \alpha_i k(x_i, x) + b \quad (13)$$

LSSVM provides better generalization performance and computational efficiency than SVM. Hence, it is more suitable for data classification forecasts.

Figure 2 reveals the operation process of the IoV's DTs model in the operation and maintenance stage, considering the actual application scenarios. DTs technology models vehicles from different angles based on historical data. Besides, it updates the model's parameters according to the new data for a more real-time and intuitive representation of the vehicle's state. Here, historical driving data is used to establish

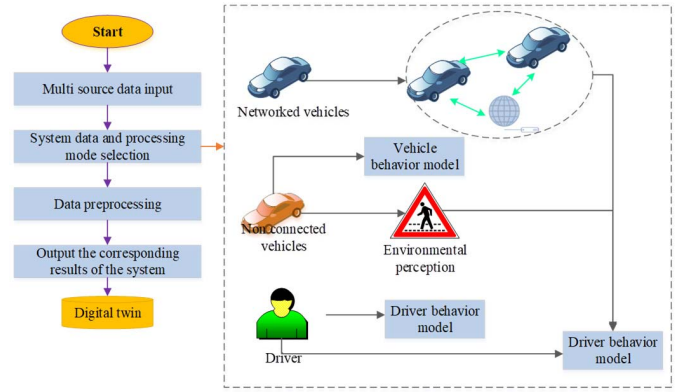


Fig. 2. IoV DTs model running process.

a driver behavior model using Machine Learning Toolbox. This model can predict the behavior of drivers under different road conditions. Meanwhile, the behavior model is shared with other vehicles to indicate the dangers, warn the driver, and improve driving safety. A DTs system based on driver behavior is established through the constant iteration of data. Then, it guides the automatic driving of actual vehicles in turn.

C. BC-Based IoV Secure Communication Framework

The BC data is inalterable and trackable. On this basis, a secure IoV communication architecture is proposed regarding the current IoV communication security issues. In IoV scenarios, critical data, such as nodes' public keys and historical communication behaviors, are stored in BC. Moreover, each BC node jointly maintains the communication security between the IoV nodes [26]. The drive test unit is regarded as the full node of BC, and the complete BC data are stored. Besides, the vehicle node is viewed as a lightweight BC node, and only the block header is stored. As per the access control logic of the smart contract, the BC node can decide the access authority of the vehicle node. In addition, it can reach a consensus on the access control results by the consensus mechanism [27].

The BC-based IoV secure communication framework comprises five modules: the BC network, access control, secure transmission protocol, and vehicle Ad Hoc modules. BC network stores information about nodes' public keys and provides an operating platform for smart contracts. Its structure is displayed in Figure 3. Nodes negotiate the session keys and encryption algorithms according to the secure transmission protocol. They also encrypt the communication data for transmission.

As IoV develops, Sybil attacks become increasingly severe. Hostile vehicles may report the tampered location information to the server, affecting the accuracy of detecting Sybil attacks. In this regard, regular location certificates issued by base stations to vehicles can prove the accuracy of the vehicle location information. Therefore, attack detection includes two steps: issuance of location certificates and location reliability assessment. A location certificate consists of the vehicle identity, base station serial number, and time stamp. Location certificates are issued by the base station based on their private

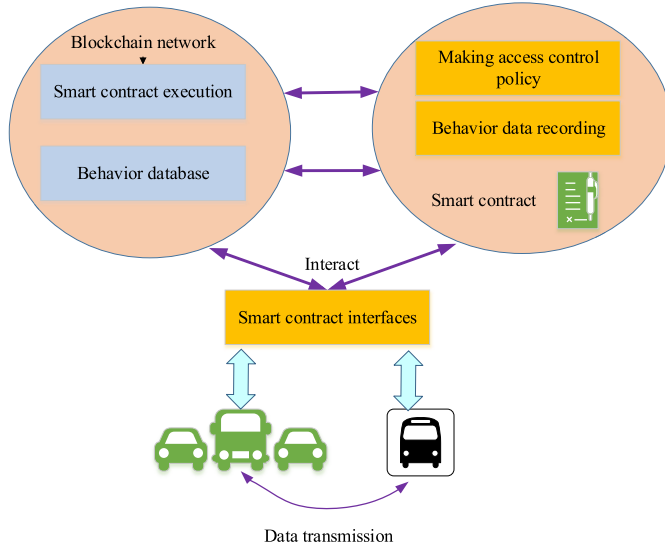


Fig. 3. BC network structure in IoV secure communication framework.

key and are difficult to forge. Afterward, subjective logic assesses the reliability of the reported information. The overall uncertainty u_i is calculated by weighting the time uncertainty u_t and the spatial uncertainty u_s :

$$u_i = \alpha \cdot u_t + (1 - \alpha) \cdot u_s \quad (14)$$

$$u_t = -e^{-\beta_t \cdot T_{cer}} + 1 \quad (15)$$

$$u_s = -e^{-\beta_s \cdot D_{cer}} + 1 \quad (16)$$

In Equations (14) ~ (16), α controls the proportion of u_t and u_s , and T_{cer} represents the difference between the time of reporting and certificate issuance. Meanwhile, D_{cer} describes the distance between the reporting location and the base station that issues the certificate. Besides, β_t and β_s are two preset parameters that control the uncertainty.

A greater time or space gap between the reported information and the authentication information indicates a higher uncertainty of the reported information. Vehicles shall apply for the latest location certificates from the nearby base stations periodically. Otherwise, the uncertainty of locations reported by the vehicles will increase gradually. Correspondingly, the expected credibility will decrease.

D. IoV Risk Forecasting Based on the Generative Adversarial Network (GAN)

IoV is a dynamic and open network with fluctuated node behaviors. The system cannot adjust its authority in time under an attack during a node's access, reducing information confidentiality [28]–[30]. GAN adopts game training to improve the prediction accuracy of the model. This method utilizes all model training samples and employs the generated data to fine-tune the model parameters. Therefore, GAN is used to build the risk forecast model of IoV nodes.

The risk forecast of IoV nodes is converted into GAN behavior sequence generation. Denote the node behavior sequence as $B = \{B_1, B_2 \dots B_n\}$, where B_i represents the behavior sequence at a particular moment. In addition,

$B_i = \{b_{i1}, b_{i2} \dots b_{im} | risk_i\}$, including m behavioral attributes, and $risk_i$ indicates the forecasted risk level.

GAN's generation model G aims to learn historical behavior data of nodes. In this way, the network can generate new behavior sequences to deceive the discriminant model. Moreover, the discriminant model D discriminates the probability that the input data is an actual sample. GAN's objective function can be described as:

$$\min_G \max_D V(D, G) = E_{b \sim P_{data(b)}} [\log D(b | risk)] + E_{z \sim P_z(z)} [\log (1 - D(G(z | risk)))] \quad (17)$$

The two models are trained alternately to improve the generation and discrimination capabilities of the model. GAN can eventually converge to the Nash equilibrium through the gaming process of alternate training. The generation model can better fit the node behavior distribution and accurately forecast the node risk [31], [32]. However, GAN trains the network through gradient descent. Consequently, the training of the GAN forecast model is unstable, reducing the prediction accuracy. The objective function of the original GAN equals the Jensen Shannon (JS) divergence. It can optimize the distribution of actual behavior data and generated behavior data of the sample. If the two distributions do not overlap, the JS divergence is usually constant, which may cause the generator gradient to vanish. Kullback Leibler (KL) divergence and JS divergence are determined according to:

$$KL(P_r \| P_g) = \int P_r(x) \log \left(\frac{P_r(x)}{P_g(x)} \right) dx \quad (18)$$

$$JS(P_r \| P_g) = \frac{1}{2} KL \left(P_r \left\| \frac{P_g(x) + P_r(x)}{2} \right\| \right) + \frac{1}{2} KL \left(P_g \left\| \frac{P_g(x) + P_r(x)}{2} \right\| \right) \quad (19)$$

Introducing Wasserstein distance can fix this problem and solve the gradient vanishing during training. Wasserstein distance compares the distribution similarity of the two behavioral data types based on histograms. Finally, a risk forecast model is proposed, namely WaGAN. WaGAN accelerates the convergence speed and reduces the training time while ensuring forecast accuracy. The Wasserstein distance measures the cost of transforming one kind of behavioral data to another type of behavioral data. Given data described by histograms, it calculates the cost of transferring one histogram to another histogram. The smaller the cost, the more similar the two. The Wasserstein distance can be written as Equation (20).

$$W(P_r, P_g) = \inf_{\gamma \in \Pi(P_r, P_g)} E_{(x,y) \sim \gamma} [\|x - y\|] \quad (20)$$

In Equation (20), $\Pi(P_r, P_g)$ represent the set of all possible joint distributions of P_r and P_g . Sampling each joint distribution can obtain a generated sample x , a real sample y , and the distance between these two samples $\|x - y\|$. Under all joint distributions γ of P_r and P_g , the lower bound of the expected sample distance is the Wasserstein distance.

In the present work, the KDDCup99 dataset is employed for simulation experiments to verify the feasibility of the proposed

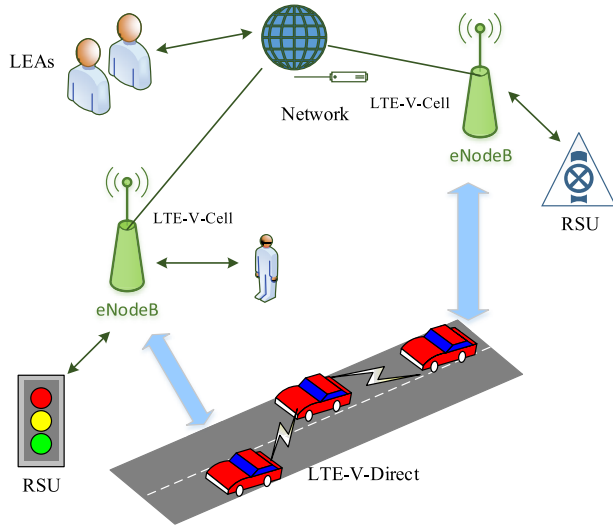


Fig. 4. LTE-V network model.

GAN-based risk forecast model. The data set is nine weeks of network connection data collected from a simulated US Air Force Local Area Network. It is divided into labeled training data and unmarked test data. The test data and training data have different probability distributions. The test data contains some attack types that do not appear in the training data, making intrusion detection more realistic. The risk forecast process is written in Python. Each connection record contains 41 fixed characteristic attributes and one class identifier. Among the 41 fixed feature attributes, nine are discrete; the others are continuous.

E. Optimization of IoV Group Access Authentication Scheme

In the Long-Term Evolution-Vehicle (LTE-V) network, mobile units need to simultaneously read various information. The data comes from different sources, such as network nodes, signs, traffic lights, and charging piles. In this time, cross-domain communication authentication is a challenge. Figure 4 signifies the LTE-V network model. The enormous mobile units are connected to the base stations in IoV. In result, there are usually network channel congestion and some security authentication problems [33], [34]. Hence, a new group access authentication scheme is proposed, namely Group Authentication and Privacy-preserving (GAP).

The proposed GAP scheme founds on the aggregation of proxy signatures and key sharing. During initialization, the user server distributes the proxy signature to the Mobile Management Entity (MME) and the Vehicle-mounted Engines (VE). During group authentication, MME and VE authenticate mutually via proxy signatures before generating a shared key. Moreover, while VE tracks the authentic identity, any trusted administrator can cooperate to complete the tracking task.

One group tries to access the LTE network at the same time. First, the leader is selected according to each VE's communication capability, storage capacity, battery status, and other characteristics. The group leader representatives of all other MME members are used for mutual authentication. After

- 1 $VE_{G1-i} \rightarrow leader: (V_{G1-i}, EID_{VE_{G1-i}}, GID, T_{G1-i})$
Calculate $w = (EID_{VE_{G1-i}} \| GID \| T_{G1-i})$ and $h_i = H_2(w)$,
send each vector to the superior;
- 2 $leader \rightarrow MME: (AUTH_{leader})$ leader receives the
signatures of all VE members, aggregate n signatures into one
signature $V_{G1} = \sum_{i=1}^n V_{G1-i}$. Calculate vector $AUTH_{leader}$ and
send to MME;
- 3 MME receives the message authentication $AUTH_{leader}$;
Check whether each T_{G1-i} timeout expires; if the timeout
expires, send the message of the verification failure to the group
leader; otherwise, enter the next verification process;
Calculate $h_i = H_2(w)$, verify the equation
$$e(V_{G1}, P) = e\left(\sum_{i=1}^n h_i H_1(EID_{VE_{G1-i}})\right);$$

After successful verification, MME selects the current time
and calculated the session key using K_{G1-i}^S MAC function;
- 4 $MME \rightarrow leader: (AUTH_{MME})$
- 5 $leader \rightarrow VE_{G1-i}: (AUTH_{MME})$, the group leader
sends an authentication message to all group members;
Verify the MME identity through the equation:
$$e(V_{MME}, P) = e(h_{MME} H_1(EID_{VE_{G1-i}}), Q);$$

If MME has a legal identity, it becomes a session key
 K_{G1-i}^S ;
- 6 Send the end confirmation message to MME to notify the MME
that the session key is successfully generated, and the
communication can be started

Fig. 5. The detailed process of the group authentication stage.

the identities of both parties are confirmed, each VE generates a session key to ensure security during MME communication. Figure 5 displays the detailed process of the group authentication stage.

Each VE in the GAP scheme receives a set of pseudo-identities and corresponding private keys from the user server during registration. The keys employed here can only be found in MME and VE and cannot be transmitted via the hollow. In this way, hostile users cannot monitor or tamper with the data. If VE's authentic identity must be revealed, the GAP scheme can provide traceability to avoid hostility and abuse of user privacy. The GAP scheme is used for VE authentication of MME. GAP is superior to other methods, such as less communication and lower calculation cost. Table II provides the details. The formal verification tool AVISPA is adopted to verify the analysis results automatically. This test aims to analyze the security of the GAP scheme and confirm that the security goals are achieved. The simulation tool is MATLAB, and the cryptographic algorithm employed is a hash function. AVISPA can simulate intruders who can receive and send messages. Once the HLPSSL specification is debugged, four checkers will check whether it encounters automatic attacks. The check results will be reported to SAFE in SUMMARY if the proposed protocol is secure. In GAP data exchange, HLPSSL is responsible for describing VE. Simulating the

TABLE II
PERFORMANCE INDICATORS OF LED TRAFFIC LIGHTS

Characteristics	AKA based on dynamic grouping	MTC access authentication	GAP
Two-way authentication	×	√	√
Anonymity	×	√	√
Anti-simulation attack	√	×	√
Unlinkability	√	√	√
Traceability	√	√	√
Anti-replay attack	√	√	√
Non-repudiation	×	×	√
Non-frame up	×	×	√
Unlinkability	√	×	√
Anti-forward attack	√	√	√
Anti-backward attack	√	√	√
Anti-internal attacks	×	×	√

intruder's attack proves that GAP can provide a secure data protection scheme.

F. Improving the Communication Protocol in IoV Data Security Transmission

SSL/TLS is widely deployed on different terminal system platforms to protect the privacy of node information. Besides, it can prevent various attacks and security threats against communication data during data transmission. The data security transmission based on the SSL/TLS protocol comprises the key agreement process and the data encryption transmission.

In the SSL/TLS protocol, data transmission between node A and node B uses a session key and a symmetric encryption algorithm for encryption and decryption. PreMasterKey uses RSA for secure transmission between communication nodes. RSA is composed of three parts: key pair generation, encryption algorithm, and decryption algorithm. M and N are taken as the two roles in cryptography. M first decomposes the plaintext into several blocks. Each block can be represented as an integer, denoted by I . M uses the public key (e, n) of N to perform operations according to Equation (20) to obtain ciphertext C . Then, the ciphertext is sent to N via the public network channel. N receives the ciphertext and uses its private key (d, n) to obtain the plaintext X according to Equation (22).

$$C = I^e \bmod n \quad (21)$$

$$C = I^e \bmod n, 1 \leq I \leq n - 1 \quad (22)$$

Given that p and q are prime numbers and conform to Equation (23).

$$e \cdot d = 1 \pmod{\phi(n)} = 1 \pmod{(p-1) \cdot (q-1)} \quad (23)$$

If I and n are relatively prime, the following can be obtained by Euler's theorem:

$$I^{\phi(n)} = I^{(p-1)(q-1)} = 1 \pmod{(n)} \quad (24)$$

$$(I^{\phi(n)})^k \cdot I \equiv I^{k\phi(n)+1} \equiv I \pmod{(n)} \quad (25)$$

If I and n are not relatively prime, the prime factorization of n can only be $n = p \times q$. Suppose that $I = hp$. In that case, I must be relatively prime to p .

According to Fermat's theorem and Euler function:

$$(I^{p-1}) \equiv 1 \pmod{p} \quad (26)$$

$$(hq)^{k\phi(n)+1} = tqp + hq = tn + tq \quad (27)$$

$$I^{k\phi(n)+1} \equiv I \pmod{n} \quad (28)$$

Similarly, the same conclusion can be obtained when $I = hp$.

The distribution law of prime numbers has not been fully explored. However, there have been some interesting conclusions. For example, the prime number theorem shows:

$$\lim_{n \rightarrow \infty} \frac{\pi(n)}{\ln(n)} = \frac{n}{\ln(n)} \quad (29)$$

In Equation (29), $\pi(n)$ represents the prime number distribution function. The probability of randomly selecting an integer as a prime number is $\frac{1}{\ln(n)}$. From a probability distribution perspective, one of $\ln(n)$ is a prime number.

It is easy to calculate d from e . Because $ed = 1 \pmod{\phi(n)}$, it can be transformed into solving a linear equation:

$$ed + (-y)\phi(n) = 1 = \gcd(e, \phi(n)) \quad (30)$$

$$\gcd(d, (p-1) \times (q-1)) = 1 \quad (31)$$

The solution can be completed using the extended Euclidean algorithm.

Since the key negotiation process in the traditional SSL/TLS protocol requires verifying the node's public key, the key negotiation will be insecure. The node certificate verification is based on asymmetric encryption and decryption, extending the verification time. Since data stored in the BC is inalterable and trackable, the public key information of the IoV node can be stored in the BC. Compared with the traditional SSL/TLS protocol, it reduces one asymmetric encryption and decryption and the verification time of the public key. A node can set the cookie's expiration time according to its storage space and the frequency of communication with other nodes. Meanwhile, it can delete cookies that have not been used for a long time and reduce storage overhead.

G. Algorithm Implementation and Simulation Experiment

The scheme' proposed here is compared with that of several existing methods in this experiment. Computational overhead and communication overhead are extremely critical performance indicators and the principal focus of this experiment. To obtain quantitative results, simulations are carried out to compare GAP and several typical authentication protocols. Assume that n vehicles form a group, which accesses the network T times. The experimental hardware is an Intel i5-750 processor with a 3.00 GHz clock frequency, and the simulation tool is Matlab 2014B.

A comparative analysis is conducted to validate the performance of the WaGAN risk forecast model reported here. Backpropagation Neural Network (BP) [35], Artificial Neural Network (ANN) [36], Deep Neural Network (DNN) [37],

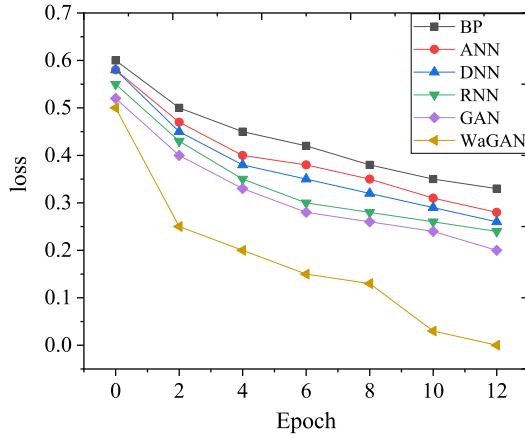


Fig. 6. Loss function variations of different risk forecast models.

Recurrent Neural Network (RNN) [38], and GAN are selected as comparative models. Moreover, a data security transmission scheme is proposed based on the Secure Sockets Layer/Transport Layer Security (SSL/TLS) protocol. It is aimed at solving the security problems of data transmission of IoV nodes. Under the same experimental conditions, the improved SSL/TLS protocol conducts key agreement with the traditional SSL/TLS, SSL, TLS, HTTP, and HTTPS protocols.

IV. RESULTS AND DISCUSSIONS

A. GAN Application in Risk Forecast

Loss function variations with iterations are summarized in Figure 6. The WaGAN model calculates the loss function using Wasserstein distance. It also collaboratively learns the probability distribution of samples using multiple generators. Hence, its learning rate gets accelerated significantly, acquiring a significant advantage at the beginning of training. After ten iterations, the loss of the WaGAN model approaches zero.

Figure 7 presents the forecast errors of the packet delivery rate given different numbers of samples in the training set. WaGAN provides the slightest forecast error on the node data packet transmission rate than other models. Given 500 pieces in the training set, WaGAN can offer a relative error of about 20.8%. Forecast errors of several risk forecast models to the node data packet transmission rate decrease as the training set enlarges. Given 5,000 samples in the dataset, WaGAN can provide a relative error of about 11.5%. Therefore, WaGAN can keep a small forecast error even under a small training set.

Figure 8 illustrates the accuracy of different risk forecast models. Given a small training set, WaGAN provides a more tremendous advantage over other models. In particular, its forecast accuracy increases by about 18% compared with BP. As the training set enlarges, all models' accuracy gets improved. WaGAN can provide a forecast accuracy of 87.5%, given 5,000 samples in the training set.

B. Performance of IoV Sybil Attack Detection Algorithm

Real vehicle movement dataset GAIA is employed to assess the proposed IoV Sybil attack detection algorithm.

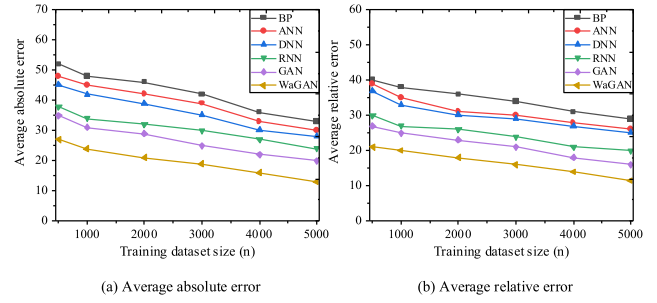


Fig. 7. Errors of different risk prediction models.

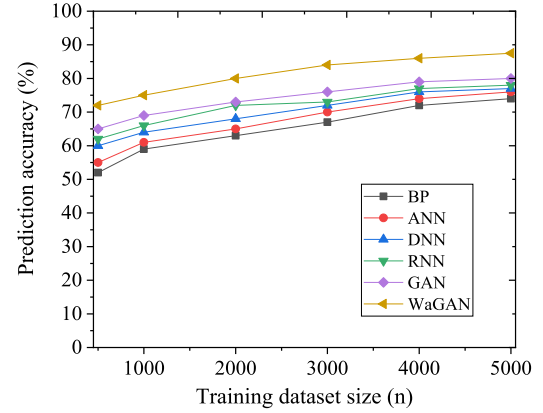


Fig. 8. Accuracy of different risk forecast models.

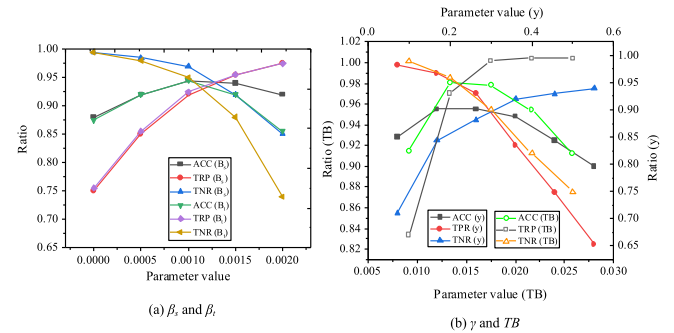


Fig. 9. Influences of parameter changes in Sybil detection algorithm on performance.

Figure 9 demonstrates the influences of β_s , β_t , γ , and Trust threshold (TB) on Accuracy (ACC), True Positive Rate (TPR), and True Negative Rate (TNR). A growth of β_s or β_t can increase TPR but reduce TNR due to the increased uncertainty of reported data. Then, wrong locations are easier to be detected. Similarly, γ and TB can affect the ACC of the model. As γ increases, the credibility of the reported information gradually increases. Consequently, it is challenging for the model to detect false locations, resulting in a decrease in TPR. The parameter TB also affects ACC. An excessive threshold may result in a situation where true data is mistaken for false data. On the contrary, a low threshold may make inaccurate information be mistaken for true data. There is an optimal TB to achieve the highest detection accuracy of the model.

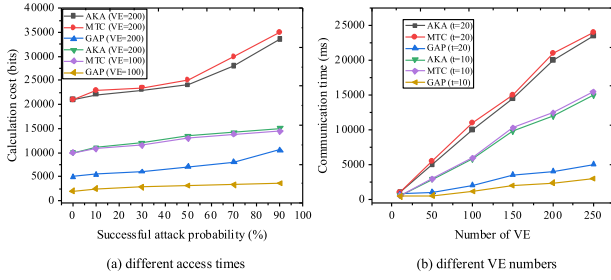


Fig. 10. Communication overhead of access authentication schemes.

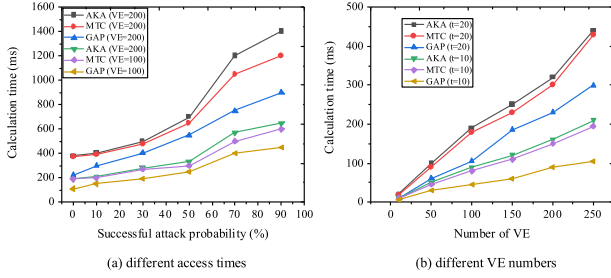


Fig. 11. Computational overhead of access authentication schemes.

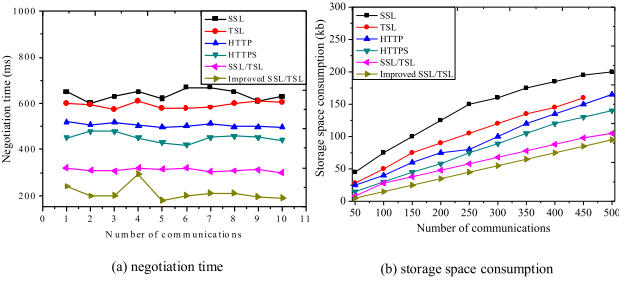


Fig. 12. Key agreement time and storage overhead of each protocol.

C. Group Authentication Results Based on the GAPL Scheme

The AVISPA tool is adopted to verify protocol security and simulate an intruder who can receive and send messages. Figure 10(a) displays the communication overhead of the three access authentication schemes. The number of accesses to the network t is set to 10 and 20, respectively. As VE increases during authentication, the GAP scheme can significantly improve the performance compared with other authentication access schemes. Once the successful attack probability reaches 60%, each scheme will consume a larger overhead.

Figure 10(b) illustrates the total computational cost of the three access authentication schemes with different VE numbers. Figure 11 presents the computational overhead of these schemes under different access times and different VE numbers. If the attack is unknown, the GAP scheme gets improved steadily below 50%, indicating a strong ability to resist unknown attacks. Since the GAP scheme practices multiple encryption algorithms, privacy can be protected via a dot multiplication operation. Hence, its overhead performance is better than others.

The GAP scheme can reduce signal exchanges between large-scale mobile units and the network, thereby diminishing

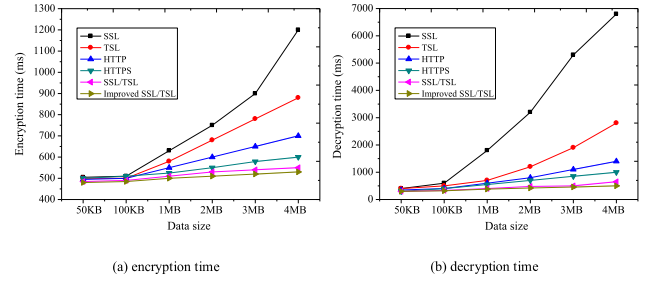


Fig. 13. Encryption and decryption time of each protocol.

the signal overload. This scheme also has powerful security features, including privacy protection and non-tampering.

D. Test Results of IoV Data Security Transmission Scheme

Simulation experiments of multiple key agreement processes between nodes analyze the experimental results statistically. The key agreement time and storage space consumption comparison of each protocol is shown in Figure 12. Traditional SSL/TLS protocol adds a random number to the session key generation process of the node. The security of the session key generation is ensured by the random number instead of the complexity of the private key itself. In this way, the asymmetric encryption and decryption process are simplified, and the efficiency of key agreement is improved. Furthermore, delay in verifying the public key certificate and the key reconstruction in the traditional SSL/TLS protocol is prolonged. Therefore, the certificate is replaced by public key information to reduce the time for public key verification.

The improved SSL/TLS protocol sets a reasonable time for cookie data, and the node deletes invalid cookie files. Thus, it requires less storage space consumption than the other five schemes. Therefore, as the number of communication nodes increases linearly, the storage overhead during the key agreement will not increase linearly but tends stable. Some storage spaces of nodes are often consumed in a scenario requiring real-time communication such as IoV. This operation can reduce the key negotiation time between nodes and improve the key agreement efficiency.

The encryption and decryption time of different protocols during data encryption transmission is tested with 50KB, 100KB, 1MB, 2MB, 3MB, and 4MB data in sequence. Figure 13 provides the comparison result of encryption and decryption time of each protocol. When the encrypted data is small, the time differences between encryption and decryption of the six protocols are minor. However, as encrypted data increases, the advantages of the improved SSL/TLS protocol are gradually manifested. In other words, the enhanced SSL/TLS protocol can present excellent time performance with a slight increase in data.

V. CONCLUSION

DTs match the current development of digital transportation; hence, the DTS-based IoV safety issues are explored. A BC-based secure communication framework is designed according

to the management requirements of 6G DTs-driven IoV. This system can detect the potential attacks of vehicle nodes while accessing information. Moreover, a BC consensus mechanism matching the IoV characteristics enhances data communication security.

The BC-based IoV secure communication framework comprises the BC network, access control, secure transmission protocol, and vehicle Ad Hoc modules. Besides, the WaGAN model is employed to construct a IoV node risk prediction model. WaGAN model adopts game training, which can fit node behavior distribution excellently and predict node risks accurately. On this basis, a new secure and efficient group access authentication scheme is proposed, namely WaGAP. Simulation experiments reveal an accelerated learning rate of the WaGAN model. This model calculates the loss function using Wasserstein distance. It also collaboratively learns the probability distribution of samples using multiple generators. The GAP scheme-based IoV group authentication provides a better communication performance than other schemes. The research results can offer a reference for advancing the intellectual development of IoV and secure communication. Nevertheless, there are some deficiencies in the present work. User privacy protection is ignored in IoV communication, which will be considered in future research.

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